

Model Card — MedTriage: Symptom-to-Department Triage Classifier

1. Model Details

Modelname	MedTriage-v1
Task	Multi-class classification — route a free-text symptom description to 1 of 11 hospital departments
Model	TF-IDF (unigrams + bigrams, 8,000 features, min_df=2, sublinear TF) + Logistic Regression (C=1, class_weight='balanced'), with temperature-scaled confidence calibration (T=2.95)
Training data	Medical Transcriptions (MTSamples, CCO public domain, mtsamples.com); 1,883 records across 11 departments after cleaning
Test data	377 held-out records (stratified 80/20 split, random_state=42); trained and evaluated on the description field only, matching exactly what the live demo receives
Preprocessing	Filtered to 11 real departments (min. 40 records each); stripped keywords naming a department to prevent label leakage
Libraries	Python 3, scikit-learn, pandas, numpy, scipy (calibration), Gradio (live demo)
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2. Intended Use

- ☐ **Intended:** Flagging the likely department for a free-text symptom description as a first-pass triage suggestion; classroom demo of an audited, calibrated NLP pipeline.
- ☐ **Out of scope:** Clinical diagnosis or real patient-care decisions; low-data departments (Pediatrics, Psychiatry) where reliability is lower; non-English or multi-symptom input.

3. Overall Performance

The model reaches 79.8% accuracy and 0.76 macro-F1 on the 377-record held-out test set, beating a majority-class baseline (19.6%) by 60.2 points, across 11 departments with natural class imbalance (Cardiology has 7x more training records than Psychiatry).

Model	Accuracy	Macro-F1	Weighted-F1
Majority-class baseline	19.6%	0.033	—
MedTriage-v1 (TF-IDF + LR, calibrated)	79.8%	0.76	0.80

Best: Ophthalmology, Cardiology, Orthopedics (F1 ≥ 0.87) — distinctive vocabulary gives strong signal. Weakest: Pediatrics (F1 = 0.39), Neurology (F1 = 0.66) — see Section 4.

4. Performance Across Text Subgroups (Bias Audit)

Key audit finding: accuracy drops by 7.8 points on short descriptions and by 9.3 points on narrative, patient-case-style text compared with terse, procedure-title-style text — exactly the writing style real patients are most likely to use.

Subgroup / style	How it was defined	Test size (n)	Accuracy
Short descriptions	≤ 15 words (median split)	~189	76.0%
Long descriptions	> 15 words (median split)	~188	83.8%
Narrative case style	e.g. “year-old”, “presents”, “history of”	104	73.1%
Terse / procedure-title style	Short clinical/procedure titles, no narrative phrasing	273	82.4%

The model was built mostly on terse clinical phrasing, so it struggles most on casual, narrative patient input. Pediatrics (F1 = 0.39) shows the same pattern — its vocabulary (“child”, “baby”) overlaps heavily with other departments.

5. Explainability Check

Method 1 — Top features (Logistic Regression coefficients): the strongest words per department are clinically specific, suggesting real medical signal rather than noise.

Cardiology: chest, coronary, pulmonary • Ophthalmology: eye, cataract, vitrectomy • Urology: bladder, prostate, inguinal • Pediatrics: child, baby, month old (generic — explains its low F1)

Method 2 — Worked misclassification example.

- ☐ Input “4-month-old with tachycardia” → predicted Pediatrics (99% confidence), though the cardiac term “tachycardia” suggests Cardiology is at least as plausible.
- ☐ Interpretation: age words (“month old”) are frequent, strongly-weighted Pediatrics features; “tachycardia” is rarer and more weakly weighted, so the generic cue dominates the specific one — the same word-matching-without-context limitation behind the Section 4 audit gap.

6. Limitations and Ethical Considerations

- ☐ **Main limitation:** The model matches words, not context — it fails when a generic word (age, setting) outweighs a specific symptom term, as shown above. Not safe for real clinical use without human oversight.
- ☐ **Data bias:** Training data is naturally imbalanced (Cardiology: 372 vs. Psychiatry: 53 records pre-split), so smaller departments are learned from far less language.
- ☐ **Label source:** Departments come from the dataset's self-reported specialty tags, which may not always reflect the clinically optimal routing.
- ☐ **Small audit sample:** The narrative-style audit subgroup has only 104 examples, so its 73.1% estimate is less precise than the larger subgroups.

7. Recommendations

- ☐ Collect more examples for low-support departments (Pediatrics, Psychiatry, Nephrology); try a contextual embedding model (e.g., clinical BERT) and re-run this audit to compare against TF-IDF; extend the existing 0.50 confidence threshold into a tiered human-review system for borderline cases.